

Bacterial Purine Nucleoside and Nucleobase Salvage: A Modular Two-Tier Recycling System

A commissioned review-style synthesis for a molecular-biology audience. Search date: 2026-08-13. Evidence base: primary literature and authoritative reviews retrieved via PubMed.

1. Executive summary

Bacteria rarely build purine nucleotides from scratch when preformed purines are available. Instead they run a compact, reusable **salvage module** organized into two tiers. In the **first tier**, purine nucleosides (adenosine, inosine, guanosine, xanthosine) are cleaved by **nucleoside phosphorylases** in a phosphate-dependent reaction that liberates the free base and **α -D-ribose 1-phosphate (R1P)**. In the **second tier**, three **type I phosphoribosyltransferases (PRTases)** reattach the freed bases to the activated sugar donor **5-phospho- α -D-ribose 1-diphosphate (PRPP)**, releasing **diphosphate (PPi)** and regenerating nucleoside monophosphates: **Apt** makes **AMP** from adenine; **HGPRT/Hpt** makes **IMP and GMP** from hypoxanthine and guanine; and **Xpt** (or the broader-specificity **Gpt** in enterobacteria) makes **XMP** from xanthine (

P 26443734

The module is best understood as a *shared phosphorolysis front end feeding three parallel, base-specific outputs*. Its second tier is ancient: the four purine PRTases are paralogues of one another and are homologous to the nucleoside phosphorylases, assembled by gene duplication and "patchwork" recruitment before the divergence of Bacteria, Archaea and Eukarya (

P 9742728

The first tier is mechanistically redundant and evolutionarily heterogeneous: purine-nucleoside phosphorylase activity is delivered by at least three unrelated protein folds — the classical trimeric/hexameric **NP-I (DeoD)** family, the **NP-II (XapA/Pdp)** family, and the more recently characterized **Cupin-fold** **PpnP**

(P 35094440

).

P 11444965

This review defines the system's boundaries, reconstructs its mechanism, surveys lineage variation, traces its deep origin, and flags the points where evidence is thin or contested.

2. Definition and biological boundaries

What is included. The salvage module comprises exactly two enzymatic operations acting on the N-glycosyl bond and the base:

1. **Phosphorolysis of purine nucleosides** to free base + R1P (PpnP; and, in most bacteria, the classical PNPs DeoD and XapA).
2. **Phosphoribosyl transfer** from PRPP onto a free purine base to give a nucleoside-5'-monophosphate + PPi (Apt, HGPRT/Hpt, Xpt/Gpt).

The defining chemistry is the making and breaking of the N-glycosyl bond that links base to ribosyl moiety, together with PRPP-dependent nucleotide formation

).

(P 26443734

By this definition the system's inputs are *extracellular or intracellular purine bases and nucleosides*, and its output is *the purine nucleotide monophosphate pool (AMP, IMP, GMP, XMP)*.

Neighboring processes that should be treated separately. Several adjacent pathways are frequently conflated with salvage but are mechanistically and genetically distinct:

- **De novo purine biosynthesis** (the *pur* regulon: PurF→...→IMP) builds IMP from PRPP, amino acids and one-carbon units. Salvage and de novo *converge* on the IMP/AMP/GMP pool but use entirely different enzymes; they are coupled only through shared PRPP demand and through regulation (Section 6).
- **Nucleotide interconversion** — IMP→AMP (PurA/PurB), IMP→XMP→GMP (GuaB/GuaA), and the deaminase/reductase steps (Add, GuaC/GuaD) — reshapes the monophosphate pool *after* salvage. These are downstream of, not part of, the salvage module, although they determine which base a given nucleoside ultimately becomes.
- **Nucleoside kinases** (e.g., guanosine–inosine kinase Gsk; adenosine kinase in some taxa) provide an *alternative, phosphorylytic-independent* route from nucleoside to nucleotide

without passing through a free base. This is a genuine competing route (Section 5), not part of the phosphorolysis→PRTase logic, and its presence varies by lineage

(
P 765747
).

- **Ribose-1-phosphate disposal** (via Deo enzymes / phosphopentomutase into central carbon metabolism) is the fate of the sugar co-product and links salvage to carbon/energy metabolism, but is not itself a salvage step.
- **Transport** (NupC/NupG nucleoside permeases; base permeases such as PbuX, PbuG, and NCS2/NCS1 family transporters; the xanthosine permease XapB) delivers substrates but is accessory to the catalytic core

(
P 30341391
).

P 11466294
).

Competing definitions. The brief treats **PpnP** as *the* purine-nucleoside phosphorylase of the module. The broader literature is more pluralistic: in *E. coli* the historically dominant purine-nucleoside phosphorylase is the *deoD*-encoded **hexameric PNP-I**, with **XapA** handling xanthosine, and PpnP is an additional broad-specificity (pyrimidine *and* purine) enzyme

(
P 17639373
).

P 11466294
).

P 35094440
).

A rigorous scope therefore treats "purine-nucleoside phosphorylase activity" as a *function* that can be discharged by any of three unrelated families, of which PpnP is the newest and most catholic member, rather than a single obligatory enzyme.

3. Mechanistic overview

Tier 1 — phosphorolysis (base liberation). The general reaction is:



The reaction is freely reversible and near-equilibrium; direction is set by substrate/product concentrations and by downstream pull. For the classical PNPs, catalysis proceeds through an oxocarbenium-ion-like transition state, exploited by transition-state-analogue inhibitors such as immucillin-H (

P 11444965

PpnP performs the same net chemistry but from a completely different scaffold: a rigid Cupin-fold dimer with a hydrophobic nucleoside pocket (

P 35094440

Tier 2 — phosphoribosyl transfer (base → nucleotide). The general reaction is:



This step is effectively **irreversible** under physiological conditions because PPi is hydrolyzed by inorganic pyrophosphatase, pulling flux toward nucleotide. The chemistry is an **in-line SN1-like nucleophilic substitution** at the ribose C1': the purine N9 attacks the anomeric carbon of PRPP with inversion of configuration ($\alpha \rightarrow \beta$), displacing PPi. Two **Mg²⁺ ions** orient and electrostatically activate the PRPP diphosphate for departure; a mobile **catalytic loop** closes over the active site as the penultimate step, sequestering the reaction from solvent (Toxoplasma HGPRT ternary complex,

P 10545171

PfHGPRT loop III',

P 27479359

Base specificity is imposed by the **C-terminal "hood"** that caps the type I PRTase core (

P 12037295

Sequence, obligation, and accessory steps.

- *Obligatory* for base salvage: at least one Tier-1 phosphorylase (or a nucleoside kinase bypass) to generate the free base, plus the matching Tier-2 PRTase.
- *Conditional*: Tier 1 is only required when the substrate arrives as a *nucleoside*; free bases enter Tier 2 directly. Conversely, if a nucleoside kinase is present, the base need never be generated at all.

- *Accessory*: transporters, ribose-1-phosphate salvage, and the interconversion enzymes that decide whether hypoxanthine-derived IMP becomes AMP or GMP.

A key ordering constraint: **a base cannot be phosphoribosylated until it is free**, so for nucleoside substrates phosphorolysis strictly precedes PRT. Because PRPP is the shared donor for both salvage and de novo synthesis, the two pathways compete for it, and PRPP availability gates Tier-2 flux.

4. Major molecular players and active assemblies

4.1 Tier 1 — nucleoside phosphorylases (three unrelated families)

Family	Representative(s)	Fold / assembly	Notes
NP-I	DeoD (PNP)	Trimeric or hexameric α/β	<i>E. coli</i> DeoD is hexameric; <i>M. tuberculosis</i> DeoD is trimeric, like mammalian PNP (P 11444965 P 17639373 P 11437593). Broad on Ino/Guo/adenosine.
NP-II	XapA (xanthosine PNP), Pdp (pyrimidine-nucleoside PNP)	Dimeric	XapA acts on xanthosine; substrate delivered by XapB permease (P 11466294).
Cupin / PpnP	PpnP	RlmC-like Cupin dimer	Broad pyrimidine+purine specificity; conserved dimer/pocket residues across bacteria (P 35094440).

The existence of three convergent folds discharging the same activity is the central surprise of Tier 1 and the reason the module is robust to loss of any one phosphorylase in many species.

4.2 Tier 2 — purine phosphoribosyltransferases (one ancestral fold, four specificities)

All are **type I PRTases**: a central five-stranded parallel β -sheet flanked by α -helices, carrying the conserved ~13-residue **PRPP-binding motif** and a variable **C-terminal hood** for the base

([P 12037295](#)

[P 9016724](#)

They act as homodimers/tetramers with two Mg^{2+} per active site.

Enzyme (gene)	Base(s)	Product(s)	Notes
Apt (adenine PRTase)	adenine	AMP	Constitutive in <i>E. coli</i> ; bacterial APRT structure and ligand-induced closure characterized in <i>Fusobacterium nucleatum</i> (P 41588988). Can also condense PRPP with the de novo intermediate AICA (P 9742728).
HGPRT/Hpt (<i>hpt</i>)	hypoxanthine, guanine	IMP, GMP	The double-specificity purine PRTase; in many organisms extends to xanthine (HGXPRT), e.g., <i>P. falciparum</i> (P 27479359).
Xpt (<i>xpt</i>)	xanthine	XMP	Dedicated xanthine PRTase in <i>B. subtilis</i> , in the <i>xpt-pbuX</i> operon (P 9098051).
Gpt (<i>gpt</i> , enterobacteria)	xanthine, guanine ($\pm$ hypoxanthine)	XMP, GMP	Broader-specificity enterobacterial enzyme; distinct locus from <i>hpt</i> (P 765747).

4.3 Accessory transporters and disposal

NupC/NupG import nucleosides
([P 30341391](#));

XapB imports xanthosine
([P 11466294](#));

base-specific permeases (PbuX for xanthine in *B. subtilis*,
([P 9098051](#));

NCS2/NCS1 families broadly) import free bases. R1P is routed by phosphopentomutase/Deo enzymes into central metabolism.

5. Evolutionary and cell-biological variation

Deep origin. The Tier-2 PRTases are **monophyletic**: adenine-, hypoxanthine-, guanine- and xanthine-PRTases are homologous to one another *and* to the nucleoside phosphorylases, and were assembled by gene duplication before the Bacteria/Archaea/Eukarya split
([P 9742728](#)).

Substrate specialization is therefore a *derived* property layered onto a promiscuous ancestral PRPP-condensing enzyme; because APRT still condenses PRPP with AICA, **adenine PRTase is arguably the best living proxy for the ancestral role**
([P 9742728](#)).

By contrast, the **Cupin-fold PpnP is a later, convergent invention** for Tier-1 chemistry
([P 35094440](#)),

and the split of the classical PNPs into hexameric vs trimeric forms is a **lineage-specific quaternary-structure elaboration**
([P 11444965](#)).

Lineage differences in the enzyme roster. - *Enterobacteria* (*E. coli*/*Salmonella*): Hpt (hypoxanthine/guanine) + Gpt (guanine/xanthine) + Apt; DeoD (hexamer) + XapA + PpnP for

Tier	1;	nucleoside	kinase	Gsk	present
------	----	------------	--------	-----	---------

(P 765747

).

P 26443734

- *Firmicutes* (*B. subtilis*): dedicated Xpt with its own permease PbuX in a purine-repressed operon, i.e., xanthine salvage is a *separately regulated, import-coupled* branch

).

P 9098051

- *Actinobacteria* (*M. tuberculosis*): trimeric DeoD PNP and a characterized HGPRT

).

P 11444965

);

P 19362594

salvage enzymes are studied as drug targets.

Alternative routes to the same outcome. - Nucleoside kinase bypass: Gsk phosphorylates guanosine/inosine directly to GMP/IMP, and adenosine kinases (where present) make AMP, *without* liberating a base — a parallel solution that skips the PRTase tier entirely

).

P 765747

- **Deamination re-routing:** adenosine deaminase (Add) converts adenosine→inosine and adenine deaminase converts adenine→hypoxanthine, shunting adenine-derived material into the HGPRT branch rather than Apt

).

P 765747

- **Redundant phosphorylases:** DeoD, XapA and PpnP overlap on several nucleosides, so no single Tier-1 enzyme is universally essential

).

P 35094440

).

P 11466294

No true compartments. Because bacteria lack organelles, the entire module is cytoplasmic; "compartmentalization" is achieved kinetically (active-site loop closure, metabolite channeling to pyrophosphatase) rather than by membranes. Physiological-state variation (starvation, nitrogen limitation) alters *expression* (e.g., xanthine dehydrogenase induction and *xpt-pbuX* repression respond to purine and nitrogen status;

).

P 9098051

far more than it alters the enzymes.

6. Constraints, dependencies, and failure modes

Ordering constraints. - For nucleoside substrates, **phosphorolysis must precede phosphoribosyl transfer** — a base cannot be salvaged until it is free. - **PRPP must be available** for Tier 2; PRPP is co-required by de novo synthesis, pyrimidine salvage, NAD/histidine/tryptophan pathways, so Tier 2 is embedded in a competitive economy for this single activated sugar. - **PPi hydrolysis** makes Tier 2 effectively irreversible and provides the thermodynamic pull that orients the whole module toward nucleotide formation.

Substrate-specificity constraints. - Apt cannot rescue *hpt/gpt* loss and vice versa: the hood dictates which base enters which nucleotide, so specificity failures create metabolically distinct phenotypes (adenine vs hypoxanthine/guanine/xanthine auxotrophy-like defects). - Whether hypoxanthine-derived **IMP** becomes **AMP** or **GMP** is decided *downstream* by the interconversion enzymes, not by salvage itself.

Regulatory integration and feedback. - The salvage **products themselves are signals**: hypoxanthine and guanine are the physiological **corepressors of PurR**, the master repressor of the *pur* regulon (holorepressor-*purF* operator $K_d \approx 3.4$ nM; aporepressor binds guanine at 1.7 μ M, hypoxanthine at 7.1 μ M)

(P 2211500

P 2089227

P 1490614

When salvage supplies ample bases, de novo synthesis is shut off — a clean example of pathway-level economy. - In Firmicutes, the same bases act largely through **cis-acting purine riboswitches** and dedicated transcription factors rather than a PurR-type protein repressor. The *xpt-pbuX* operon is repressed up to 160-fold by hypoxanthine+guanine (

P 9098051

a guanine-sensing riboswitch (e.g., the *B. subtilis* *yxjA* purine riboswitch) downregulates a nucleoside transporter by cotranscriptional RNA strand exchange that favors a transcriptional terminator

(P 35348734

and dedicated regulators such as NupR govern nucleoside permease expression

(
P 35862946
).

The regulatory *outcome* (throttle uptake and salvage when purines are abundant) is conserved, but the *mechanism* differs from enterobacterial PurR — a caution against generalizing salvage regulation from *E. coli* to all bacteria.

Failure modes and their exploitation. - Loss of HGPRT/PNP salvage sensitizes or desensitizes cells to **purine-analogue prodrugs** (6-mercaptopurine, 6-thioguanine), which must be salvaged to become toxic nucleotides; these analogues also activate PurR

(
P 2089227
).

This is the basis of *gpt/hpt*-based positive/negative genetic selections

(
P 1761220
).

- In pathogens (*M. tuberculosis*, apicomplexan parasites) the salvage enzymes are pursued as **drug targets**, exploiting transition-state analogues (immucillins) and species-specific active-site features

(
P 11444965

P 19362594

P 27479359
).

What is ruled out. The chemistry forbids certain "shortcuts": a PRTase cannot act on an intact nucleoside (no free N9), and a phosphorylase cannot on its own produce a nucleotide (it makes a base + R1P, not an NMP). Thus any path from an imported nucleoside to an NMP must either (i) go base→PRPP via a PRTase, or (ii) be phosphorylated by a kinase; there is no single-enzyme phosphorolytic route to a nucleotide.

7. Controversies and open questions

1. Is PpnP a principal salvage phosphorylase or a specialist? PpnP was defined biochemically and structurally only recently

(
P 35094440
);

its *quantitative* contribution to purine salvage relative to DeoD and XapA in vivo, and across species, is not well established. The brief's framing of PpnP as the module's phosphorylase is a simplification the field has not fully validated.

2. **Boundary between salvage and de novo.** Because APRT can condense PRPP with AICA (P 9742728),

the historical line between "salvage" and "biosynthesis" is blurry at the level of enzyme capability. How often such promiscuous activities matter physiologically is unresolved.

3. **Generalization across organisms.** Much mechanistic detail (two-Mg²⁺ catalysis, loop closure, hood-based specificity) comes from *eukaryotic parasite* HGPRTs (Toxoplasma, Plasmodium;

P 10545171

P 27479359

and is *assumed* to transfer to bacterial enzymes. It very likely does, given the shared fold, but direct bacterial structural coverage — especially of Hpt, Xpt and Gpt with bound PRPP·Mg²⁺ and base — remains comparatively thin. Bacterial APRT structures are only now appearing

P 41588988

4. **Quaternary-structure significance.** Why *E. coli* DeoD is hexameric while mycobacterial and mammalian PNPases are trimeric

P 11444965

— and whether this affects cooperativity, substrate range, or inhibitor design — is not fully explained.

5. **Redundancy and essentiality.** With three unrelated phosphorylase folds and a kinase bypass, which Tier-1 activities are genuinely essential under nutrient limitation, and how flux partitions among them, is largely inferred rather than measured by flux analysis.

6. **Transporter–enzyme coupling.** The degree to which base/nucleoside uptake is kinetically coupled to (or rate-limiting for) salvage — e.g., XapB→XapA→Gpt for xanthosine — is suggested by operon organization

(
P 9098051

P 11466294
)

but not quantified in most systems.

Overarching caution. The salvage "module" is a useful abstraction, but its enzyme roster, regulation, and even its dominant phosphorylase differ between enterobacteria, Firmicutes and Actinobacteria. Claims should be anchored to the organism in which they were demonstrated and not assumed universal.

8. Key references

- Jensen KF, Dandanell G, Hove-Jensen B, Willemoës M. **Nucleotides, Nucleosides, and Nucleobases.** *EcoSal Plus* (2008).

P 26443734

— Authoritative review of *E. coli*/*Salmonella* nucleotide metabolism, including salvage and transport.

- Becerra A, Lazcano A. **The role of gene duplication in the evolution of purine nucleotide salvage pathways.** *Orig Life Evol Biosph* (1998).

P 9742728

— Monophyly of purine PRTases and their homology to phosphorylases; pre-LUCA-era assembly by patchwork.

- Wen L, et al. **Crystal structures of a new class of pyrimidine/purine nucleoside phosphorylase revealed a Cupin fold.** (2022).

P 35094440

— Structural definition of PpnP as a distinct NP class.

- Basso LA, et al. **Purine nucleoside phosphorylase from *M. tuberculosis*...** (2001).

P 11444965

— Trimeric vs hexameric DeoD; transition-state inhibition.

- Modrak-Wójcik A, et al. **Ionization of the phosphate cosubstrate on phosphorolysis by PNP (*E. coli* and human).** (2008).

P 17639373

— *deoD* PNP-I kinetics/cooperativity.

- Kadziola A, Neuhard J, Larsen S. **Product-bound *B. caldolyticus* UPRTase...** (2002).

P 12037295

— Type I PRTase fold, PRPP motif, C-terminal hood.

- Héroux A, et al. **Toxoplasma HGPRT ternary complex with XMP·PPi·2 Mg²⁺.** (1999).

P 10545171

— Two-Mg²⁺ activation mechanism of purine PRTases.

- Roy S, et al. **W181 and loop III' in *P. falciparum* HGXPRT.** (2016).

P 27479359

— Catalytic-loop closure and base specificity.

- Kim et al. **Ligand-induced conformational changes in APRT from *Fusobacterium nucleatum*.** (2025).

P 41588988

— Bacterial adenine PRTase structure.

- Christiansen LC, et al. **Xanthine metabolism in *B. subtilis*: *xpt-pbuX* operon.** (1997).

P 9098051

— Dedicated Xpt + xanthine permease; purine/ nitrogen-controlled regulation.

- Jochimsen B, Nygaard P, Vestergaard T. **Location of *add*, *gsk*, *hpt*.** (1975).

P 765747

— Genetic separability of salvage/deaminase/kinase loci.

- Rolfes RJ, Zalkin H. **Purification of the *E. coli* pur repressor and corepressors.** (1990).

P 2211500

Meng LM, Nygaard P (1990)

P 2089227

Steiert JG, et al. (1992)

P 1490614

— Hypoxanthine/guanine as PurR corepressors.

- Nørholm MHH, Dandanell G. **Specificity and topology of the *E. coli* xanthosine permease XapB.** (2001).

P 11466294

— Substrate delivery to XapA.

- Almagro G, et al. **NupC/NupG nucleoside transporters.** (2018).

P 30341391

- Cheng L, et al. **Cotranscriptional RNA strand exchange underlies the gene regulation mechanism in a purine-sensing transcriptional riboswitch.** (2022).

P 35348734

— *B. subtilis* yxjA guanine riboswitch controlling nucleoside transport.

- Qin Y, et al. **NupR is a nucleoside permease regulator in *Bacillus thuringiensis*.** (2022).

P 35862946

— Protein-level control of nucleoside uptake in a Firmicute.

Uncertainty statement: Structural/mechanistic detail is strongest for the type I PRTase chemistry (well supported across multiple organisms) and for the PpnP fold; it is weakest for direct bacterial Hpt/Xpt/Gpt structures with full substrate complements and for in vivo flux partitioning among redundant Tier-1 enzymes. Where findings derive from eukaryotic parasites or from a single species, this is stated in-text and should not be generalized to all bacteria.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery